
Kinematics for Two-Body Nuclear Reaction in Python

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Abstract

We present a computational tool for calculating and visualising kinematic curves in nuclear reaction experiments, enabling rigorous comparison between measured data and theoretical predictions. The Python-based software simulates non-relativistic (low energy) two-body reaction kinematics $A(a,b)B$ by solving energy and momentum conservation equations in laboratory frame, incorporating Q -values. Users can generate kinematic lines for projectile-target interactions and optionally include detector-specific effects such as detector geometry and angular acceptance. The tool provides an efficient interface for experimental validation and maintains accessibility for researchers with limited computational expertise. Designed for nuclear physics applications, it facilitates experimental design, on-the-fly kinematic verification, and quantitative model comparisons.

Keywords: nuclear reaction, low-energy, two-body, kinematic, Python, computation

1. Introduction

In the middle ages in Europe, a group of pseudo-scientists called Alchemists claimed that they could transform baser metals like iron, copper, etc. into the expensive metals like gold and silver. Though there was no scientific basis for such a claim at that time, the discovery of radioactivity rekindled this idea of transforming one metal into another in the scientific minds again. From our understanding of the atomic nucleus, we can see that changing the neutron or proton number or both can bring about a change in the nuclear structure which will change the element itself.

To transform the nucleus artificially, a large amount of energy (a few MeV) needs to be provided, at least equal to the binding energy. This energy can be supplied by introducing a nuclear particle in the nucleus. This process of supplying a nuclear particle to the nucleus of a given atom to bring about a transformation in the atom is called a nuclear reaction.

1.1 Present Tools

Nuclear reaction studies and radioactive beam experiments rely heavily on computational tools to predict reaction kinematics, cross-sections, and beam properties. Two widely used software packages that predict the kinematics of reactions are:

1. NRV (Nuclear Reaction Video): NRV solves relativistic two-body kinematics, providing energy-angle correlations for reaction products. has the provision for classifying the reactions in different types: fusion, transfer, elastic, etc. It takes one set of input parameters for those reactions and returns some of the other parameters. It also plots the kinematic line for one given nuclear reaction given its corresponding initial energy and information about the target and projectile.

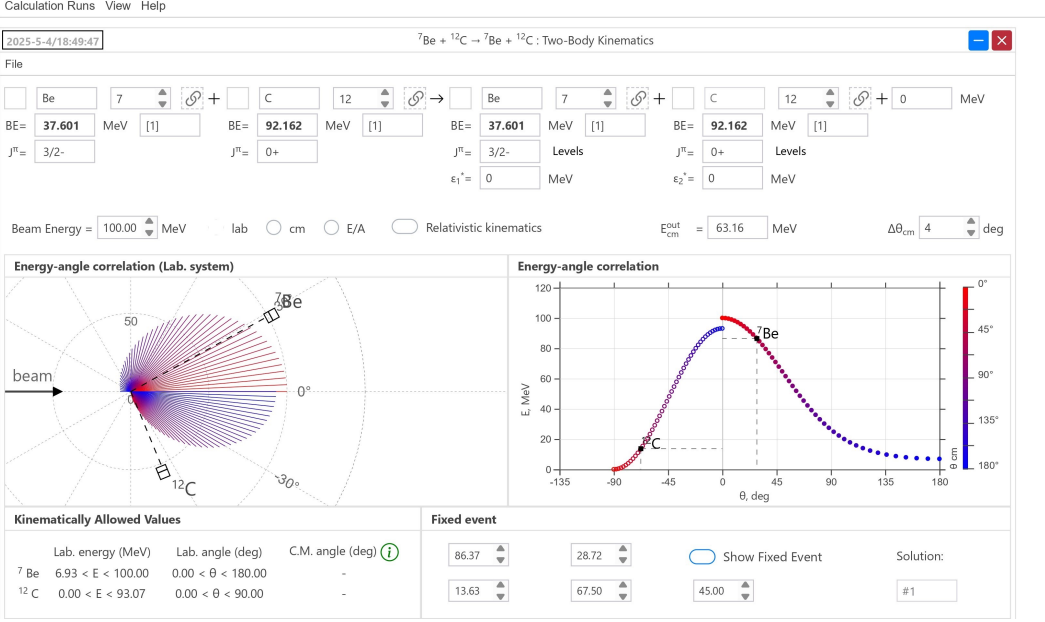


Figure 1. Two body Kinematics using NRv

2. LISE++ : It is a complete simulation software for designing and analysing experiments with radioactive ion beams (RIBs), particularly at fragmentation facilities. It can be used to plot reaction kinematics out of a lot of other utilities as well. It considers target and detector thickness and hence calculates energy losses as well.

1.2 Why a new code?

We have devised a new python code for kinematics of nuclear reaction for the following reasons:

- Even though NRv works with relativistic kinematics, it **does not consider energy loss** in the reaction due to target thickness. Neither can it plot **multiple kinematic lines** in one graph.
- NRv does consider any information about the **detector system**.
- LISE++ does work with energy loss and detector, but cannot plot more than one reaction in the same graph and hence makes it difficult to compare plots.
- While traditional nuclear physics computations have predominantly relied on C++ for its computational efficiency, the field is increasingly adopting Python due to its advantages in rapid development and scientific workflow integration. This paradigm shift motivated our implementation of the two-body nuclear reaction kinematics script in Python.

2. Theory

The kinematic relationship between the outgoing particle energy (E_b) and reaction angle (θ_b) for the nuclear reaction $A(a,b)B$ is derived from fundamental conservation laws.

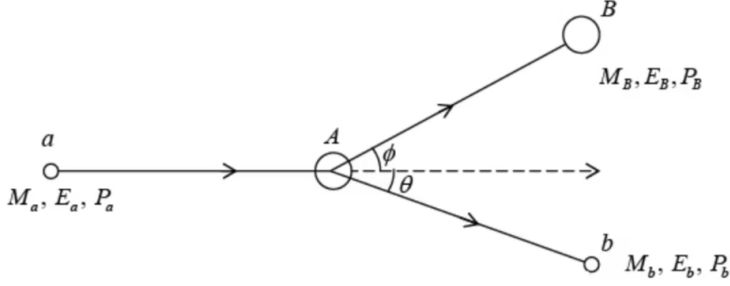


Figure 2. Schematic of two body nuclear reaction

From the conservation of energy, the Q-value of the reaction is given by:

$$Q = E_b + E_B - E_a$$

From the conservation of momentum,

$$\begin{aligned} p_a &= p_b \cos(\theta_b) + p_B \cos(\theta_B) \\ 0 &= p_b \sin(\theta_b) - p_B \sin(\theta_B) \end{aligned}$$

So,

$$\begin{aligned} \sqrt{2m_a E_a} &= \sqrt{2m_b E_b} \cos(\theta_b) + \sqrt{2m_B E_B} \cos(\theta_B) \\ 0 &= \sqrt{2m_b E_b} \sin(\theta_b) - \sqrt{2m_B E_B} \sin(\theta_B) \end{aligned}$$

Using these equations, we can arrive at the relation between E_b and θ_b :

$$\sqrt{E_b} = \frac{\sqrt{m_a m_b E_a} \cos \theta_b \pm \sqrt{m_a m_b E_a \cos^2 \theta_b + (m_b + m_B)(m_B Q + (m_B - m_a)E_a)}}{m_b + m_B}$$

This is the equation we use to find the kinematic of the lighter particle b. Our code uses this equation of the lighter particle to make the kinematic plots, given the input energy of the projectile and the atomic masses of the reacting species. Next, we integrate this with the thickness of the target block and hence introduce the energy loss in the reaction. The rate at which a particle loses energy per unit path length is known as the stopping power of the medium. We use Bethe Bloch's equation for that:

$$-\frac{dE}{dx} = K z^2 \frac{4\pi Z \rho N_A}{A} \left[\ln \left(\frac{2m_e v^2}{I} \right) - \ln(1 - \beta^2) - \beta^2 \right]$$

where:

- $K = 9 \times 10^9$, a constant
- z = projectile atomic mass
- Z = target atomic mass
- ρ = target density
- N_A = Avogadro number
- A = target atomic mass
- I = mean excitation mass
- $\beta = \frac{v}{c}$, the relativistic term

This is the relativistic version of the equation. We mainly deal with low-energy reactions for which we shall adopt the non-relativistic version in our calculations. Thus putting $\beta = 0$, we get:

$$-\left\langle \frac{dE}{dx} \right\rangle = Kz^2 \frac{4\pi Z\rho N_A}{A} \left[\ln \left(\frac{2m_e v^2}{I} \right) \right]$$

This will get the rate of energy loss in the target as well as the detector.

Modern nuclear experiments combine multiple detector technologies to achieve comprehensive measurements of reaction products. The choice of detectors depends on the specific physics goals, balancing resolution, efficiency, and rate capability. These detectors are designed to identify particles, determine their energies, and measure angular distributions. Some of the commonly used detectors are: Silicon Detectors (SSD, DSSD), Gas Detectors (Ionization Chambers, TPCs), Scintillation Detectors, Sodium iodide or lanthanum bromide as Gamma-Ray Detectors and Boron Trifluoride as Neutron Detectors. To calculate the total energy loss, we use Bragg's formula:

$$\left\langle \frac{dE}{dx} \right\rangle_{\text{compound}} = \sum_i w_i \left\langle \frac{dE}{dx} \right\rangle_i$$

where

- w_i is the fraction by weight of each atom in the compound.
- $\left\langle \frac{dE}{dx} \right\rangle_i$ is the energy loss by each atom.

3. Methods and Features

We have used Python in developing the entire code because of its versatility and wide range of applicability. Numpy, Random, Pandas and Matplotlib libraries have been used widely. The code takes the number of counts as user input, along with the rest of the information as follows:

- atomic number of projectile
- atomic number of target
- atomic mass of projectile
- atomic mass of target
- atomic mass of lighter product
- atomic mass of heavier product
- atomic number of lighter product
- density of target in kg/m^3
- ionisation potential of target in keV
- radius of target in m
- input energy of projectile in keV
- thickness of target in m

The inputs are to be given in the exact order as above for each of the reactions to be plotted. For each reaction, the inputs are entered in one excel file. The user shall create as many excel files as needed for the desired number of reactions.

3.1 General flow of the code:

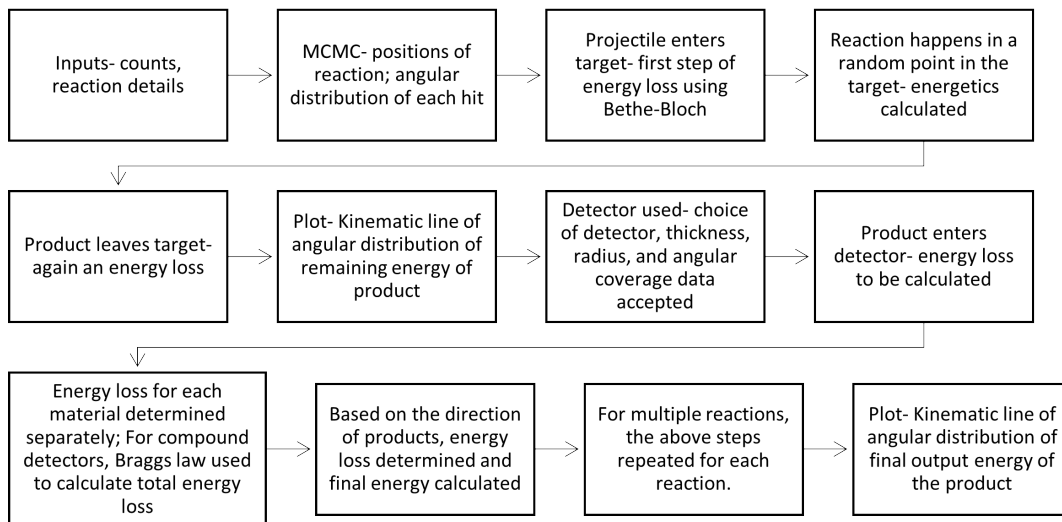


Figure 3. Flow of the code

3.2 Inputs

The data of one file is read through pandas. Correspondingly other necessary constants are also defined. If multiple reactions are to be plotted, this input block from the code has to be executed multiple times. The details are defined in the README file.

3.3 Monte Carlo Method

Given some initial energy to the projectile, it will hit the target and penetrate some distance. We use Monte Carlo (MC) simulation to determine the distance moved by the projectile inside the target before the reaction happens. As the projectile moves this distance, some energy is lost. So the remaining energy is available to the projectile for the reaction. Now, after the reaction, the product can move in any direction. We again use MC to determine a range of probable angular distributions for each hit.

3.4 Inside the target

As the projectile penetrates the target, some energy is lost. We calculate this energy by the Bethe-Bloch equation for non-relativistic particles. Now, we calculate the energetics of the reaction and thus get an energy matrix for the products. This is the entire set of energy values for each hit/event and each angular direction of products. Further the products will travel out of the target and hence lose more energy in the path. Calculating the remaining distance matrix, from the position and angle of reaction in the target, we can obtain the energy lost by the particles to move this distance. Thus this gives us the remaining energy matrix.

3.5 Adding detector

Adding a detector to this setup is optional. In this version of code, the user can choose three different types of detector materials- Silicon, Caesium Iodide and Sodium Iodide. Again, the detector width as well as angular coverage is defined in an input excel file. Using Bragg's equation and some known constants, we calculate the energy each particle will deposit while passing through the detector. The

distance travelled by the particles inside the detector is calculated by its thickness and the direction of travel of the particle. The remaining energy matrix is thus calculated after calculating the energy loss in the detectors.

3.6 Multiple reactions

If the user wants to plot two reactions, indicated blocks for "Reaction 2:" has to be uncommented and used. For more number of reactions, blocks of "Reaction 2:" has to be copied and pastes separate in cells. The variable names has to be changed as well and then the code will be ready to run. More details are mentioned in README.

3.7 Plotting

There are two plotting stages. When the product leaves the target, we can create a plot (or multiplots) to get the kinematic line for the energy distribution of the particles before the detector is placed. Again, we can get another and final plot after the detector is used. If working with multiple reactions, all reaction cells have to be executed before plotting. Consequently, relevant lines have to be added for plotting each reaction as well in the final cell before executing.

4. Code Testing & Results

We have used two reactions to demonstrate the functioning of the code: $^{12}\text{C}(^7\text{Be}, ^7\text{Be})^{12}\text{C}$ and $^{12}\text{C}(^7\text{Be}, ^3\text{He})^{16}\text{O}$. The inputs for reaction are as follows:

Table 1. Table for input data

Reaction	Z (t)	A (pj)	A (t)	A (lp)	A (hp)	Z (lp)	ρ (t)	IP (t)	Rad (t)	IE (pj)	Thick (t)
C(Be,Be)C	6	7	12	7	12	4	2267	0.011	1.5×10^{-05}	35000	1.5×10^{-05}
C(Be, He)O	6	7	12	3	16	2	2267	0.011	1.5×10^{-05}	35000	1.5×10^{-05}

After the product emerged from the target:

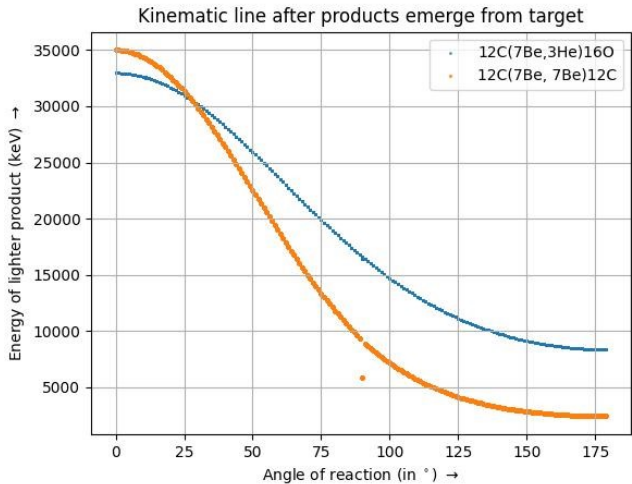


Figure 4. Energy distribution of products of the given reactions after the products have left the target

Now we use a detector of CsI having a thickness of 20 μm . We use it only for the reaction $^{12}\text{C}(^7\text{Be}, ^3\text{He})^{16}\text{O}$. The final energy profile is as follows:

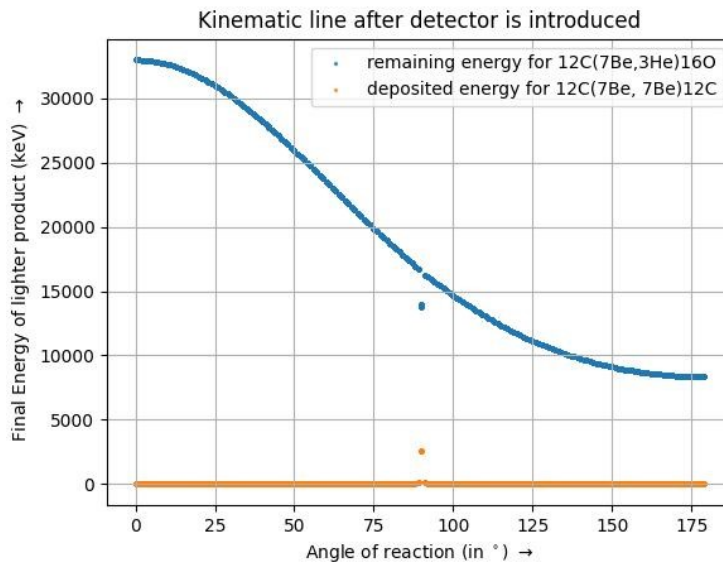


Figure 5. Energy profile of reaction after adding detector

5. Future Prospects

This code models two body nuclear reactions taking into account energy losses from target and detector information. In future, we would like to extend this to three body reactions. Also, we want to generalise the multiple reaction provision and make it more user-friendly. Finally, we can also try to reduce the time complexity of the entire code.

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Data Availability Statement This code can be accessed from GitHub in the given URL: <https://github.com/Sanchari-Sen/Nuclear-Kinematics.git>

To access this code, download and run it as a .ipynb file. Further instructions have been specified in the README file.

Ethical Standards This research is based on original work, with completely original ideas. The user is free to download and run this code and even try to implement the changes mentioned in

the future prospects. However, all changes and utilities must be made with proper referencing and citation of this original work.

Appendix 1. Code for the kinematic calculation

The code is available on GitHub at the mentioned URL. It has two parts: with and without a detector.

References

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